

PROJECT REPORT

AWS DevOps Intern Assessment

Submitted In Fulfillment of the DevOps Internship Assignment Tasks

Submission Date: 09 - July - 2026

Project Details:

- **Platform:** Amazon Web Services (AWS) Cloud
- **Operating System:** Ubuntu Server (22.04 LTS / 24.04 LTS)
- **Web Server:** Nginx
- **Containerization:** Docker

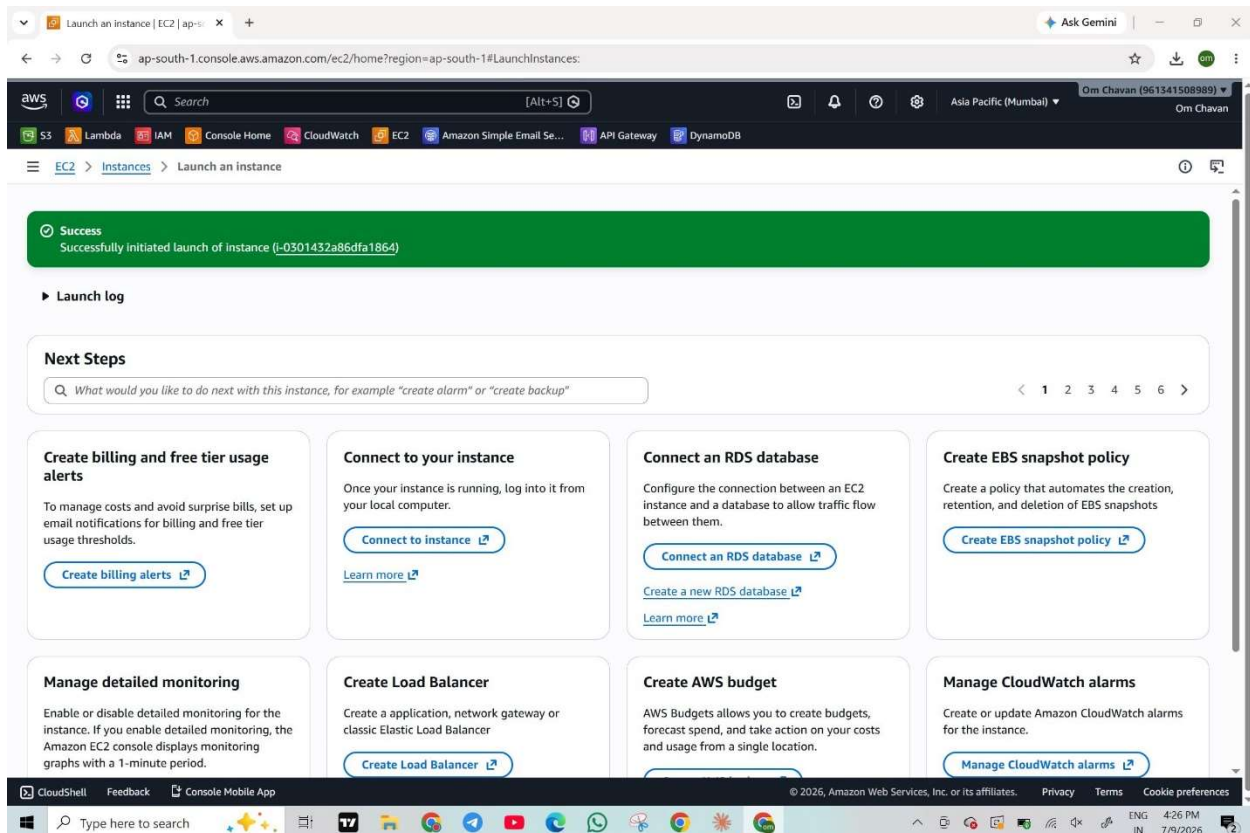
Prepared By:

- **Name:** Om Mohan Chavan
- **Qualification:** Masters in Computer Science
 - **Email:** chavhanom039@gmail.com
 - **Mobile no.:-** +91 9146051314

Task 1: Create an AWS EC2 Instance

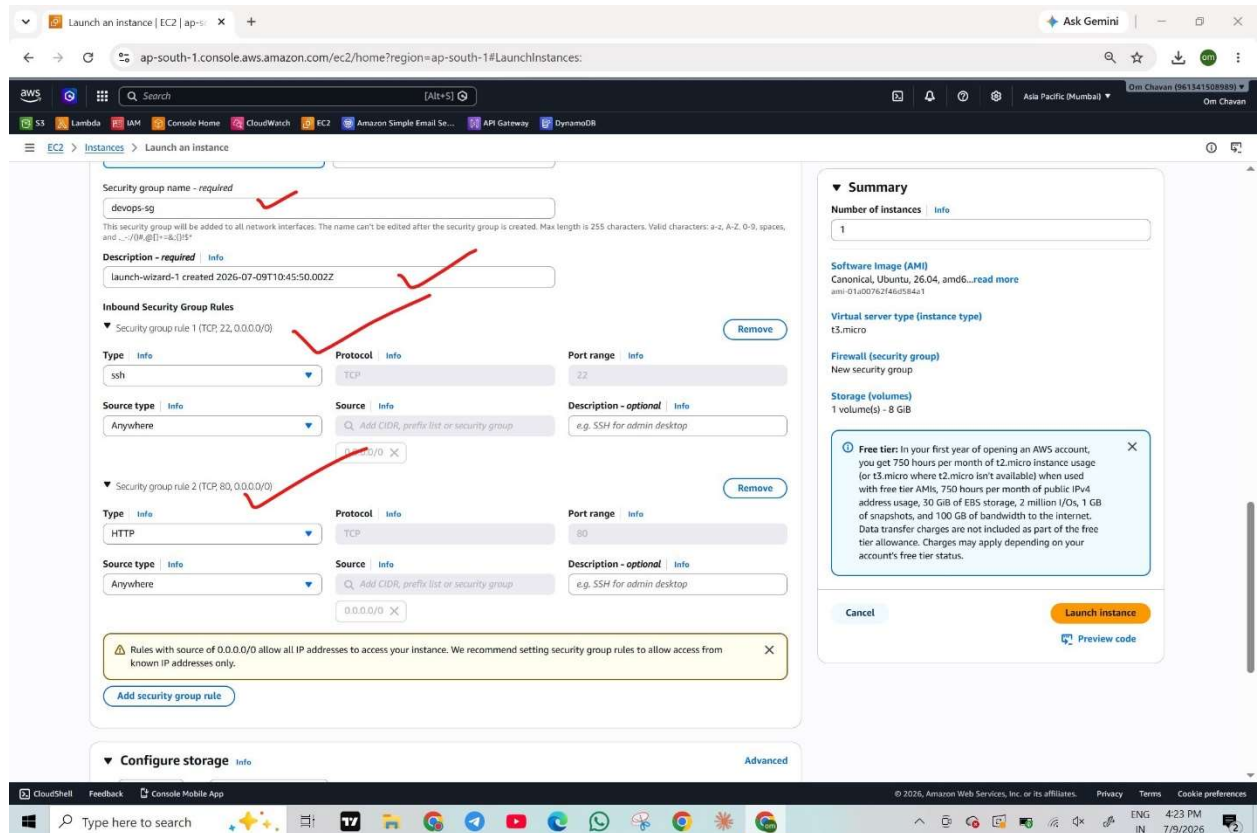
Step 1.1: Launch the Instance

1. Log in to your **AWS Management Console**.
2. Navigate to **EC2** and click **Launch instance**.
3. **Name:** DevOps-Intern-Assessment
4. **Application and OS Images (AMI):** Select **Ubuntu** (choose the *Ubuntu Server 24.04 LTS* or *22.04 LTS - Free Tier Eligible*).
5. **Instance type:** `t2.micro` (Free tier eligible).
6. **Key pair:** Click **Create new key pair**.
 - o Name it `devops-key`.
 - o Private key file format: Select `.pem` (for Mac/Linux or Windows OpenSSH) or `.ppk` (for PuTTY).
 - o Download and save it securely.



Step 1.2: Configure Network & Security Group

- Under **Network settings**, check the following boxes:
 - Allow SSH traffic from** -> Select *Anywhere* (0.0.0.0/0) or *My IP* (safer).
 - Allow HTTP traffic from the internet** (This opens port 80).
- Click **Launch instance**.
-



Step 1.3: Connect via SSH

Open your terminal (or Git Bash on Windows) and run:

Bash

```
# 1. Navigate to where you downloaded your key  
cd /path/to/your/key
```

```
# 2. Change permissions of the key file (Required for Linux/Mac)  
chmod 400 devops-key.pem
```

```
# 3. Connect to your instance (Get the Public IP from your EC2 dashboard)  
ssh -i "devops-key.pem" ubuntu@<YOUR_EC2_PUBLIC_IP>
```

Task 2: Linux Basics & Nginx Setup

Once logged into your EC2 instance via SSH, execute the following commands:

Step 2.1: Update Packages & Install Nginx

Bash

```
# Update the package lists
sudo apt update && sudo apt upgrade -y
```

```
# Install Nginx web server
sudo apt install nginx -y
```

Step 2.2: Check and Restart Nginx

Bash

```
# Check status
sudo systemctl status nginx
```

```
# Restart Nginx
sudo systemctl restart nginx
```

Step 2.3: Check System Metrics (Take screenshots of these)

Bash

```
# Check disk usage
df -h
```

```
# Check memory (RAM) usage
free -m
```

```
# Check running processes
top -n 1
```

Instances | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Instances:

Search [Alt+S]

Asia Pacific (Mumbai) Om Chavan (961541508989) Om Chavan

EC2 > Instances

Instances (1) Info

Last updated less than a minute ago

Connect Instance state Actions Launch Instances

Save filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	devops-intern...	i-0301432a86dfa1864	Running	t3.micro	3/3 checks passed	ap-south-1b	ec2-43-204-143-79.ap-...

Select an instance

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Type here to search

Instances | EC2 | ap-south-1 Welcome to nginx!

Not secure 43.204.143.79

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Type here to search

5:34 PM 7/9/2026

Task 3: Host a Simple Website

Step 3.1: Create your Custom HTML File

We will replace the default Nginx landing page with your personal assignment details.

Bash

```
# Navigate to the web root directory
cd /var/www/html

# Backup the original index page
sudo mv index.nginx-debian.html index.nginx-debian.html.bak

# Create a new index.html file
sudo nano index.html
```

Step 3.2: Add Content

Paste the following HTML template into the editor (replace the placeholder text with your actual details):

HTML

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>My DevOps Assignment</title>

  <style>

    body { font-family: Arial, sans-serif; text-align: center; margin-top:
80px; background:#f4f4f4; }

    h1 { color: #2c3e50; }

    p { font-size: 18px; color: #34495e; }

  </style>

</head>

<body>

  <h1>Hello, I'm OM MOHAN CHAVAN</h1>
```

```
<p><b>College:</b> VISHVLATA COLLAGE NASHIK</p>
```

```
<p><b>Branch:</b> COMPUTER SCIENCE ]</p>
```

```
<p><b>Email:</b> chavhanom039@gmail.com</p>
```

```
<p><b>Date:</b> <span id="date"></span></p>
```

```
<script>
```

```
    document.getElementById("date").innerText = new Date().toLocaleString();
```

```
</script>
```

```
</body>
```

```
</html>
```

Step 3.3: Verify

Open your browser and enter your **EC2 Public IP** address (http://<YOUR_EC2_PUBLIC_IP>). You should see your custom web page.

```
aws
Search [Alt+S]
SS Lambda IAM Console Home CloudWatch EC2 Amazon Simple Email Se... API Gateway DynamoDB
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System information as of Thu Jul 9 13:22:23 UTC 2026

System load: 0.0 Temperature: -273.1 C
Usage of /: 35.9% of 6.61GB Processes: 133
Memory usage: 33% Users logged in: 0
Swap usage: 0% IPV4 address for ens5: 172.31.3.198

Expanded Security Maintenance for Applications is not enabled.

59 updates can be applied immediately.
47 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 9 13:05:00 2026 from 103.226.29.113
ubuntu@ip-172-31-3-198:~$ cd /var/www/html
ubuntu@ip-172-31-3-198:/var/www/html$ pwd
/var/www/html
ubuntu@ip-172-31-3-198:/var/www/html$ ls -la
total 24
drwxr-xr-x 2 root root 4096 Jul 9 13:13 .
drwxr-xr-x 3 root root 4096 Jul 9 11:58 ..
-rw-r--r-- 1 root root 1024 Jul 9 13:00 index.html.swp
-rw-r--r-- 1 root root 615 Jul 9 11:58 index.html
-rw-r--r-- 1 root root 0 Jul 9 12:18 index.nginx-debian.html.save
-rw-r--r-- 1 root root 649 Jul 9 13:12 index.nginx-debian.html
-rw-r--r-- 1 root root 649 Jul 9 12:55 intex.html
ubuntu@ip-172-31-3-198:/var/www/html$ sudo nano index.html
ubuntu@ip-172-31-3-198:/var/www/html$ sudo rm -f index.nginx-debian.html

i-0301432a86dfa1864 (devops-intern-server)
PublicIPs: 43.204.143.79 PrivateIPs: 172.31.3.198
```

```
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Type here to search
Instances | EC2 | ap-south-1 | EC2 Instance Connect | ap-south-1 | Welcome to nginx | Ask Gemini | Asia Pacific (Mumbai) | Om Chavan (961341508989) | Om Chavan

aws
Search [Alt+S]
SS Lambda IAM Console Home CloudWatch EC2 Amazon Simple Email Se... API Gateway DynamoDB
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 9 13:05:00 2026 from 103.226.29.113
ubuntu@ip-172-31-3-198:~$ cd /var/www/html
ubuntu@ip-172-31-3-198:/var/www/html$ pwd
/var/www/html
ubuntu@ip-172-31-3-198:/var/www/html$ ls -la
total 24
drwxr-xr-x 2 root root 4096 Jul 9 13:13 .
drwxr-xr-x 3 root root 4096 Jul 9 11:58 ..
-rw-r--r-- 1 root root 1024 Jul 9 13:00 index.html.swp
-rw-r--r-- 1 root root 615 Jul 9 11:58 index.html
-rw-r--r-- 1 root root 0 Jul 9 12:18 index.nginx-debian.html.save
-rw-r--r-- 1 root root 649 Jul 9 13:12 index.nginx-debian.html
-rw-r--r-- 1 root root 649 Jul 9 12:55 intex.html
ubuntu@ip-172-31-3-198:/var/www/html$ sudo nano index.html
ubuntu@ip-172-31-3-198:/var/www/html$ sudo rm -f index.nginx-debian.html
ubuntu@ip-172-31-3-198:/var/www/html$ sudo systemctl restart nginx
ubuntu@ip-172-31-3-198:/var/www/html$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-09 13:29:30 UTC; 32s ago
   Invocation: b4089e6f025f4baab7a06ea95f127d4de
     Docs: man:nginx(8)
   Process: 5557 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 5560 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 5561 (nginx)
     Tasks: 3 (limit: 627)
    Memory: 3M (peak: 3.1M)
       CPU: 28ms
   CGroup: /system.slice/nginx.service
           └─5561 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─5562 "nginx: worker process"
               └─5563 "nginx: worker process"

Jul 09 13:29:30 ip-172-31-3-198 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jul 09 13:29:30 ip-172-31-3-198 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-3-198:/var/www/html$ q

i-0301432a86dfa1864 (devops-intern-server)
PublicIPs: 43.204.143.79 PrivateIPs: 172.31.3.198
```


Instances | EC2 | ap-south-1

EC2 Instance Connect | ap-south-1

My DevOps Assignment

Ask Gemini

ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connType=standard&instanceId=i-0301432a86dfa1864&osUser=ubuntu&sshPort=22&addressFamily=...

aws

Search [Alt+S]

Asia Pacific (Mumbai)

Om Chavan (961341508989)

Om Chavan

S3 Lambda IAM Console Home CloudWatch EC2 Amazon Simple Email Service API Gateway DynamoDB

Tasks: 3 (limit: 627)
Memory: 3M (peak: 3.1M)
CPU: 28ms
CGroup: /system.slice/nginx.service

- 5561 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
- 5562 "nginx: worker process"
- 5563 "nginx: worker process"

Jul 09 13:29:30 ip-172-31-3-198 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jul 09 13:29:30 ip-172-31-3-198 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-3-198:/var/www/html\$ q
Command 'q' not found, but can be installed with:
sudo apt install python3-q-text-as-data
ubuntu@ip-172-31-3-198:/var/www/html\$ q
q: command not found
ubuntu@ip-172-31-3-198:/var/www/html\$ curl http://localhost
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <title>My DevOps Assignment</title>
 <style>
 body { font-family: Arial, sans-serif; text-align: center; margin-top: 80px; background:#f4f4f4; }
 h1 { color: #2c3e50; }
 p { font-size: 18px; color: #34495e; }
 </style>
</head>
<body>
 <h1>Hello, I'm OM MOHAN CHAVAN</h1>
 <p>College: VISHVLATA COLLAGE NASHIK</p>
 <p>Branch: COMPUTER SCIENCE]</p>
 <p>Email: chavhanom039@gmail.com</p>
 <p>Date: </p>
 <script>
 document.getElementById("date").innerText = new Date().toLocaleDateString();
 </script>
</body>
</html>
ubuntu@ip-172-31-3-198:/var/www/html\$

i-0301432a86dfa1864 (devops-intern-server)
PublicIPs: 43.204.143.79 PrivateIPs: 172.31.3.198

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Instances | EC2 | ap-south-1 | EC2 Instance Connect | ap-south-1 | Dashboard | EC2 | ap-south-1 | My DevOps Assignment | omchavan039/devops-intern | Ask Gemini

Not secure 43.204.143.79

Hello, I'm OM MOHAN CHAVAN

College: VISHVLATA COLLAGE NASHIK

Branch: COMPUTER SCIENCE

Email: chavhanom039@gmail.com

Date: Thu Jul 09 2026

Type here to search

7:22 PM 7/9/2026

Bonus Tasks

Since the assignment lists a few options, here are the guides for **Bonus Task A (Attach an Elastic IP)** and **Bonus Task B (Install Docker and run hello-world)** to maximize your credit.

Bonus A: Attach an Elastic IP

1. Go to your **AWS Console** -> **EC2 Dashboard**.
2. On the left menu under *Network & Security*, click **Elastic IPs**.
3. Click **Allocate Elastic IP address**, leave defaults, and click **Allocate**.
4. Select your newly created Elastic IP, click **Actions** -> **Associate Elastic IP address**.
5. Select your **Instance** from the list and click **Associate**. *(Note: Use this new IP to access your website and update your GitHub documentation if needed).*

Instances | EC2 | ap-south-1 | EC2 Instance Connect | ap-south-1 | Dashboard | EC2 | ap-south-1 | My DevOps Assignment | omchavan039/devops-inter | Ask Gemini

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Home:

aws Search [Alt+S] Asia Pacific (Mumbai) Om Chavan (961341508989) Om Chavan

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EC2

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▼ **Instances**

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager [New](#)

▼ **Images**

AMIs

AMI Catalog

▼ **Elastic Block Store**

Volumes

Snapshots

Lifecycle Manager

▼ **Network & Security**

Security Groups

Resources

You are using the following Amazon EC2 resources in the Asia Pacific (Mumbai) Region:

Instances (running)	1	Auto Scaling Groups	0	Capacity Reservations	0
Dedicated Hosts	0	Elastic IPs	0	Instances	1
Key pairs	1	Load balancers	0	Placement groups	0
Security groups	2	Snapshots	0	Volumes	1

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

[Launch instance](#)

[Migrate a server](#)

Note: Your instances will launch in the Asia Pacific (Mumbai) Region

Service health

[AWS Health Dashboard](#)

Region: Asia Pacific (Mumbai)

Status: ✔ This service is operating normally.

Zones

Zone name	Zone ID
ap-south-1a	aps1-az1
ap-south-1b	aps1-az3
ap-south-1c	aps1-az2

Account attributes

Default VPC

vpc-00c5dc3ba1f98662c

Settings

Data protection and security

Allowed AMIs

Zones

EC2 Serial Console

Default credit specification

Managed resources

EC2 console preferences

Explore AWS

10 Things You Can Do Today to Reduce AWS Costs

Explore how to effectively manage your AWS costs without compromising on performance or capacity. [Learn more](#)

Optimize EC2 Cost with Spot Instances and EC2 Auto Scaling

Get started with EC2 Spot Instances, EC2 Auto Scaling, and Launch Templates by following this step by step tutorial. [Learn more](#)

Save up to 90% on EC2 with Spot Instances

Optimize price-performance by combining EC2 purchase options in a single EC2 ASG. [Learn more](#)

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https://us-east-1.console.aws.amazon.com/iam/home?region=ap-south-1

Type here to search

Instances | EC2 | ap-south-1 | EC2 Instance Connect | ap-south-1 | Allocate Elastic IP address | My DevOps Assignment | omchavan039/devops-inter | Ask Gemini

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#AllocateAddress:

aws Search [Alt+S] Asia Pacific (Mumbai) Om Chavan (961341508989) Om Chavan

S3 Lambda IAM Console Home CloudWatch EC2 Amazon Simple Email Service API Gateway DynamoDB

EC2 > **Elastic IP addresses** > **Allocate Elastic IP address**

Allocate Elastic IP address [info](#)

Elastic IP address settings [info](#)

Public IPv4 address pool

- ☒ Amazon's pool of IPv4 addresses
- ☐ Public IPv4 address that you bring to your AWS account with BYOIP. (option disabled because no pools found) [Learn more](#)
- ☐ Customer-owned pool of IPv4 addresses created from your on-premises network for use with an Outpost. (option disabled because no customer owned pools found) [Learn more](#)
- ☐ Allocate using an IPv4 IPAM pool (option disabled because no public IPv4 IPAM pools with AWS service as EC2 were found)

Network border group [info](#)

ap-south-1

Global static IP addresses

AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. [Learn more](#)

[Create accelerator](#)

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tag

[Cancel](#) [Allocate](#)

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Instances | EC2 | ap-south-1 | EC2 Instance Connect | ap-south-1 | Elastic IP addresses | EC2 | My DevOps Assignment | omchavan039/devops-inter | Ask Gemini

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Addresses:PublicIp=13.204.32.19

aws

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EC2 > Elastic IP addresses

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Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager [New](#)

▼ Images

AMIs
AMI Catalog

▼ Elastic Block Store

Volumes
Snapshots
Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IP address allocated successfully.
Elastic IP address 13.204.32.19 [Associate this Elastic IP address](#)

Elastic IP addresses (1) [Info](#)

Find elastic IP addresses by attribute or tag

Public IPv4 address: 13.204.32.19 [Clear filters](#)

☐	Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record
☐	-	13.204.32.19	Public IP	eipalloc-01703872029a061c4	-

Select an elastic IP address

[View IP address usage and recommendations to release unused IPs with Public IP insights](#)

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Instances | EC2 | ap-south-1 | EC2 Instance Connect | ap-south-1 | Associate Elastic IP address | My DevOps Assignment | omchavan039/devops-inter | Ask Gemini

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#AssociateAddress:PublicIp=13.204.32.19

aws

Search [Alt+S]

S3 Lambda IAM Console Home CloudWatch EC2 Amazon Simple Email Service API Gateway DynamoDB

EC2 > Elastic IP addresses > Associate Elastic IP address

Associate Elastic IP address [Info](#)

Choose the instance or network interface to associate to this Elastic IP address (13.204.32.19)

Elastic IP address: 13.204.32.19

Resource type
Choose the type of resource with which to associate the Elastic IP address.

☒ Instance
☐ Network interface

⚠️ If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. [Learn more](#)

If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.

Instance

Private IP address
The private IP address with which to associate the Elastic IP address.

Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.
☒ Allow this Elastic IP address to be reassociated

[Cancel](#) [Associate](#)

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Instances | EC2 | ap-south-1 x EC2 Instance Connect | ap-south-1 x Elastic IP addresses | EC2 | x My DevOps Assignment x omchavan039/devops-inter x + Ask Gemini

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Instances:

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S3 Lambda IAM Console Home CloudWatch EC2 Amazon Simple Email Service API Gateway DynamoDB

EC2 > Instances

EC2

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▼ Elastic Block Store

Volumes
Snapshots
Lifecycle Manager

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Security Groups

Instances (1) Info

Last updated less than a minute ago

Connect Instance state Actions Launch Instances

Saved filter sets
Choose filter set Find Instance by attribute or tag (case-sensitive)

Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Sec
3/3 checks passed	ap-south-1b	ec2-13-204-32-19.ap-s...	13.204.32.19	13.204.32.19	-	disabled	dev

Select an instance

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Instances | EC2 | ap-south-1 x EC2 Instance Connect | ap-south-1 x Elastic IP addresses | EC2 | x My DevOps Assignment x My DevOps Assignment x omchavan039/devops-inter x + Ask Gemini

Not secure 13.204.32.19

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Branch: COMPUTER SCIENCE]

Email: chavhanom039@gmail.com

Date: Thu Jul 09 2026

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7:29 PM 7/9/2026

Bonus B: Install Docker and Run Hello-World

Back in your EC2 terminal terminal, execute:

Bash

```
# Install Docker
sudo apt install docker.io -y

# Start and enable Docker service
sudo systemctl start docker
sudo systemctl enable docker

# Run the hello-world container to verify
sudo docker run hello-world
```

Deliverable Screenshot Reminder: Take a screenshot of the terminal output showing the "Hello from Docker!" success message.

